

Beating the odds in the innovation arena: The role of market and technology signals classification and noise

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Abstract

Innovation is a risky business. Notwithstanding the attention in research on the determinants of innovation success, the estimates of the extremely high failure rates (ranging from 40% to 90%) have not come down. As a result, new approaches have to be found to address the issues that prevent organizations from reaching innovation success over a longer period of time. In this study, Signal Detection Theory (SDT) is introduced to show that it may be less important to improve innovation practices in companies, than it is to change the nature of the projects that enter the corporate innovation funnel. In our empirical study of 44 innovation projects in the pharmaceutical and electronics industry, we show that pursuing more noisy signals (uncertain technological and market opportunities) is currently beneficial for innovation success. In general, companies allocate too few resources to noisy innovation opportunities. Interestingly, we find a positive, and not an inverted U-shaped, relationship between noise/signal ratios and innovation success. This indicates that there are almost no companies that take too much risk in the current competitive environment and that many can gain from increasing their noise tolerance. The implications from this study are quite counter-intuitive in a sense that 'risk' is not the cause of the innovation problems, but may be a solution.

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1. Introduction

Innovations are the life-blood of most firms (Shane, 2000; Wind & Mahajan, 1988). Yet, while it appears that many companies only develop some successful innovations by accident, a few other firms seem to have a consistent supply of rewarding innovations (Crawford & Di Benedetto, 2003). For instance, the Minnesota Mining and Manufacturing Company (3M) is widely recognized as a truly successful and innovative company. Today, 3M makes 66,000 products as diverse as fire hose linings, surgical gowns, bingo supplies and Scotch tape (3M Annual Report 2003). The company consistently ranks in the top 10 of Fortune's list of most admired companies and makes more than a quarter of its sales

from products less than 4 years old. In the biotechnology industry, companies like Amgen have become industry leaders with a consistent flow of new products such as *Epogen*, while other companies in the industry are taken over by others because they have no new products to replace their older ones, which are running out of patent protection.

The potential causes for innovation performance differences have been widely established in earlier research, such as: variations in the quality of market analysis, differences in product design capabilities, timing, and marketing efforts (Griffin & Hauser, 1996; Henard & Szymanski, 2001; Ottum & Moore, 1997). From a strategic marketing perspective, many companies seem to have trouble accumulating and exploiting scarce innovation resources. Scientists adhering to the resource-based views argue that companies have to *acquire* and *leverage* scarce and difficult to imitate resources and capabilities to establish a sustainable competitive advantage (Barney, 1991; Day, 1994; McEvily, Eisenhardt, & Prescott, 2004). Interestingly, while much research is available and knowledge grows

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